FacePix — AI API & Monthly Operating Cost Report

Flutter (Android / iOS / Web) • Firebase backend • fal.ai generation APIs • AdMob rewarded-ad credit economy

Prepared: 12 August 2026 | **Scope:** recurring monthly running costs only. **Excluded on request:** Google Play Console and Apple Developer Program registration fees.
Prices verified: August 2026 public rate cards (fal.ai, Google AI / Gemini API, Firebase). All figures in USD. Providers are pay-as-you-go — no monthly minimum, no subscription.

1. Recommended stack — which API for which feature

Feature	Recommended model / API	Provider	Unit cost	Why this one
Face check (pre-validation)	**google_mlkit_face_detection**	**On-device (Flutter)**	**$0.0000**	**Runs locally on the phone. Blocks bad uploads *before* any paid call. Zero API cost.**
Avatars (anime / cartoon / character)	FLUX.2 [flash] (image-to-image)	fal.ai	$0.005 / image	$0.005 per megapixel, sub-second. Best cost/quality for stylised output where exact likeness matters less.
Avatars — budget tier	FLUX.1 [schnell]	fal.ai	$0.003 / image	$0.003/MP — cheapest usable image model. Use for free-tier / ad-funded users.
Professional portraits (LinkedIn headshots)	FLUX.2 [pro] edit (multi-reference, identity-preserving)	fal.ai	$0.030 / image	$0.03/MP. Multi-reference editing keeps the user's actual face — essential for headshots.
Portraits — premium tier	Gemini 3 Pro Image (“Nano Banana Pro”)	Google AI	$0.067 / image	1,120 output tokens @ $60 / 1M tokens for a 1K image. Best prompt-following and text rendering; use for paid users only.
ID / passport photos (plain background + crop)	BiRefNet v2 (cut-out) + FLUX.2 [flash] (retouch)	fal.ai	$0.007 / image	BiRefNet bills ~$0.0008/compute-second (≈$0.002 per image); retouch $0.005. Cropping/resizing is done on-device for free.
ID photos — compliance tier	Bria RMBG 2.0 + FLUX.2 [flash]	fal.ai	$0.023 / image	$0.018 per inference. Trained only on licensed data — use if you need commercial/legal indemnity.
Videos (photo → 5-second clip)	**LTX-2 Fast (image-to-video, 1080p)**	**fal.ai**	**$0.200 / 5s clip**	**$0.04 per second of output. Cheapest credible 1080p image-to-video today.**
Video — alternates	Wan 2.5 / LTX-2 Pro / Kling 2.5 Turbo Pro / Kling 3.0 Pro	fal.ai	$0.25 / $0.30 $0.35 / $0.56	$0.05 / $0.06 / $0.07 / $0.112 per second respectively, for a 5-second clip.

Billing units — read this carefully.

- **Image models on fal.ai are billed per megapixel of output, rounded up to the next full megapixel** — not per token. A 1024×1024 image = 1 MP = 1 billing unit. A 512×512 image still rounds up to 1 MP, so **always generate at 1024×1024**; going smaller saves you nothing.
- **Video models are billed per second of generated video**, and the per-second rate rises with resolution (LTX-2 Fast: $0.04/s at 1080p, $0.08/s at 1440p, $0.16/s at 2160p). Lock your app to 1080p and 5 seconds.
- **Only Gemini is token-billed:** 560 tokens per input image, 1,120 output tokens for a 1K image, at $60 per 1M image-output tokens = $0.067. Google's Batch API halves this to ~$0.034 if you accept up to 24-hour turnaround (good for a “we'll email your headshots” flow, not for live generation).

2. Backend / infrastructure costs (Firebase)

Service	Free allowance (no charge)	Rate after free tier
Authentication	50,000 monthly active users	Google Identity Platform rates. **Phone/SMS OTP is billed per message from day one** — prefer Google / Apple / email sign-in.
Cloud Firestore	1 GiB stored, 50,000 reads/day, 20,000 writes/day, 20,000 deletes/day	~$0.06 per 100K reads, ~$0.18 per 100K writes, ~$0.18/GiB stored per month
Cloud Storage (user photos & videos)	5 GB stored, 100 GB/month download, 50,000 download ops/month	~$0.026/GB-month standard storage, ~$0.12/GB network egress
Cloud Functions (API proxy — see §6)	2,000,000 invocations/month	$0.40 per million invocations

Storage is the sneaky one: one 1024×1024 PNG ≈ 1.5 MB and one 5-second 1080p clip ≈ 3 MB. At 1,000 active users you generate roughly **9 GB per month**. Auto-delete generated media after 30 days and Firebase stays in the single-digit dollars for a long time.

3. Cost of one average user, per month

Assumed behaviour of one active user (typical for an ad-supported credit app): **5 images + 0.5 videos per month** (i.e. one in two users makes one video).

What the user does	Volume / month	Unit cost	Cost to you
Sign-up, login, session	unlimited	$0.0000	$0.000
Face check before every upload	unlimited	$0.0000 (on-device)	$0.000
Avatars (60% of image mix)	3 images	$0.005	$0.015
Professional portraits (25%)	1.25 images	$0.030	$0.038
ID / passport photos (15%)	0.75 images	$0.007	$0.005
Video clips (5s, 1080p)	0.5 videos	$0.200	$0.100
Firebase storage, egress, database	~12 MB	—	$0.005
Total cost of one average user	**5 images + 0.5 video**		**≈ $0.163 / month**

Blended image cost = $0.012 per image. Round the per-user figure to **$0.16 – $0.18 per active user per month** for planning. Videos are 61% of that number while being only 9% of the generations.

4. Projected total monthly bill by user base

Stage	Active users	Images / mo	Videos / mo	Image cost	Video cost	Firebase	Total / month
A — Launch & testing	100	500	50	$6.00	$10.00	$0.00	**≈ $16**
B — Early growth	1,000	5,000	500	$60.00	$100.00	$3.00	**≈ $163**
C — Scale	5,000	25,000	2,500	$300.00	$500.00	$25.00	**≈ $825**
D — Established	25,000	125,000	12,500	$1,500.00	$2,500.00	$150.00	**≈ $4,150**

Stage A is the honest answer to "what will this cost me at launch?": about $16–20 a month, all in. Add ~10% headroom for retries, failed generations and users who re-roll a result they don't like — so budget **$20/month at launch** and **$180/month at 1,000 users**.

> **Cost scales almost perfectly linearly with generations, not with users.** There is no fixed platform fee anywhere in this stack. If nobody generates anything this month, your bill is $0.00 plus a few cents of storage.

5. Credit economy — will rewarded ads actually pay for this?

Rewarded-video eCPM (revenue per 1,000 ad views) in 2026, and what one ad view is worth to you:

User region	Rewarded eCPM	Revenue per 1 ad view	Verdict against your costs
Tier 1 — US, UK, Japan, Australia	$15 – $40	$0.015 – $0.040	One ad comfortably funds one image ($0.012). Even a portrait ($0.03) roughly breaks even.
Tier 2 — EU, Gulf, Korea	$6 – $15	$0.006 – $0.015	One ad funds one cheap avatar. Portraits need 2–3 ads.
Tier 3 — India, Pakistan, SE Asia, LATAM, Africa	$1 – $4	$0.001 – $0.004	One ad does **not** cover one image. You need 3–5 ads per image here.

> **The video problem.** One 5-second video costs you **$0.20**. That is **13 ad views in the US** and **50–200 ad views in Tier-3 markets**. If "1 rewarded ad = 1 credit = 1 video", you lose roughly $0.19 on every single video a Tier-3 user makes. At 500 videos/month that is a $95 hole with nothing coming back.

Recommended credit pricing

Action	Your cost	Charge the user	Ads needed to break even (Tier 3)
Avatar (FLUX.2 flash)	$0.005	1 credit	2 – 5 ads
ID / passport photo	$0.007	1 credit	2 – 7 ads
Professional portrait	$0.030	2 credits	8 – 30 ads
5-second video	$0.200	**10 credits, or paid IAP only**	50 – 200 ads

Practical rule: **1 rewarded ad = 1 credit; 1 credit = 1 image; video = 10 credits or an in-app purchase.** Give 3–5 free credits on install so a new user reaches the "wow" moment without watching an ad first — that costs you about $0.05 per install and is the cheapest retention you will ever buy.

6. Ten decisions that keep this bill small

- **Never call a paid API without an on-device face check first.** ML Kit is free and kills 100% of the “user uploaded a picture of their dog” spend.
- **Never ship API keys in the Flutter app.** Put a Cloud Function (or your own tiny server) in front of fal.ai. A leaked key in a decompiled APK becomes someone else's unlimited video budget on your card. Cloud Functions gives you 2M free invocations — this costs you nothing and is non-negotiable.
- **Set a hard spend cap in the fal.ai dashboard and a Google Cloud budget alert** at, say, $50 and $200. Do this on day one.
- **Rate-limit per user server-side** (e.g. max 20 generations/day) so one abusive account can't burn $100 overnight.
- **Generate at exactly 1024×1024.** Billing rounds up to the whole megapixel, so anything smaller is wasted money.
- **Lock video to 5 seconds at 1080p.** 1440p doubles and 2160p quadruples your per-second rate.
- **Cache aggressively.** Same photo + same style = return the stored result instead of regenerating. Users re-tap far more than you expect.
- **Auto-delete generated media from Cloud Storage after 30 days** (lifecycle rule) and keep only a thumbnail. This is what stops storage from quietly becoming your biggest line item at scale.
- **Route by tier:** free/ad users → FLUX.1 schnell or FLUX.2 flash; paying users → FLUX.2 pro or Gemini 3 Pro Image. Same app, 6× cost difference.
- **Use Gemini's Batch API (50% off) for anything non-instant**, such as a “10 professional headshots, ready in a few hours” pack.

7. One-line summary

FacePix costs **≈ $0.16 per active user per month** — about **$16–20/month at 100 users**, **$163/month at 1,000 users**, and **$825/month at 5,000 users**, with no fixed platform fees. Images are cheap enough that rewarded ads pay for them; **video is 60% of your cost and must be priced at 10 credits or sold as an in-app purchase**, or it will be the only thing that ever makes this app lose money.

Sources (rate cards checked August 2026): fal.ai model pricing (FLUX.1 schnell $0.003/MP, FLUX.2 flash $0.005/MP, FLUX.2 pro $0.03/MP, LTX-2 Fast $0.04/s 1080p, LTX-2 Pro $0.06/s, Wan 2.5 $0.05/s, Kling 2.5 Turbo Pro $0.07/s, Kling 3.0 Pro $0.112/s, Bria RMBG 2.0 $0.018/inference, BiRefNet v2 $0.0008/compute-second) · Google Gemini Developer API pricing (image output $60 / 1M tokens; 1,120 tokens per 1K image = $0.067; Batch API –50%) · Firebase pricing page (Blaze free tiers and rates) · 2026 mobile ad benchmarks for rewarded-video eCPM.
Provider prices change; re-check fal.ai and the Gemini pricing page before you finalise your credit pricing.